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United States Patent Application Publication

20250276745

Kind Code

A1

Publication Date

September 04, 2025

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VEHICLE ACCESSORY

Abstract

A vehicle accessory includes a body portion. The body portion is made substantially of aluminum composite material. The vehicle accessory further includes at least one connector for selectively attaching the body portion to a portion of an automobile. The body portion includes a front edge and a rear edge. The front edge is curved along at least a portion thereof. The front and rear edges are laterally opposed from each other. The rear edge extends substantially linearly from a first corner to a second corner. The first and second corners are formed at junctions between the front edge and the rear edge.

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Appl. No.: 19/190236

Filed: April 25, 2025

Related U.S. Application Data

parent US continuation 18228581 20230731 parent-grant-document US 12286165 child US 19190236

parent US continuation 17105798 20201127 parent-grant-document US 11713087 child US 18228581

us-provisional-application US 62941373 20191127

Publication Classification

Int. Cl.: B62D35/00 (20060101); B60R19/48 (20060101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application is a continuation of U.S. patent application Ser. No. 17/105,798, filed on Nov. 27, 2020, which claims priority to U.S. Provisional Application Ser. No. 62/941,373, filed Nov. 27, 2019, the disclosures of which are expressly incorporated herein by reference.

BACKGROUND AND SUMMARY OF THE DISCLOSURE

[0002] The present disclosure relates to a vehicle accessory. More specifically, the present disclosure relates to a vehicle accessory made of aluminum composite material that is both lightweight and durable, thus comprising a resilient yet functional accessory able to increase the aerodynamic performance of a vehicle. Additionally, the present disclosure relates to a vehicle accessory that increases the aesthetic appearance of a vehicle.

[0003] Vehicle accessories that increase the aerodynamic performance of a vehicle are regularly implemented in high performance and recreational vehicles alike. Specifically, vehicle accessories may, among others, increase the fuel efficiency of a vehicle, the comfort of a driver or passenger while driving, and the aerodynamic effects of a vehicle. Additionally, vehicle accessories can increase the aesthetic appearance of a vehicle.

[0004] Illustrative types of vehicle accessories include, but are not limited to, front splitters, canards, side skirts, and diffusers. First, front splitters increase front-end downforce by sending the high-pressure air over the top of the vehicle, and the low-pressure air under the vehicle, thereby increasing overall vehicle traction and improving cornering and grip. Second, canards increase front-end downforce by deflecting the high-pressure air around the wheel well, where otherwise, it would become trapped, thereby increasing overall vehicle traction and improving cornering and grip as well. Third, side skirts increase overall downforce by lowering pressure under the car-speeding up air flow or removing the air all together, once more increasing vehicle traction and grip. Finally, diffusers tailor airflow from a high-speed dynamic under the vehicle to a much tamer ambient surrounding airflow, thereby reducing drag and increasing grip in the vehicle. All vehicle accessories noted also can increase the aesthetic appearance of the vehicle as well.

[0005] However, conventional vehicle accessories typically have two main flaws. First, conventional vehicle accessories made from metals, though being resilient to damage, increase the overall weight of the vehicle, leading to decreased fuel and aerodynamic efficiency. Thus, vehicle accessories made from metal sacrifice vehicle performance for resistance to heat. Additionally, metal vehicle accessories can decrease the aesthetics of the vehicle, in comparison to other materials.

[0006] Second, conventional vehicle accessories made from carbon fiber or the like can increase the overall performance of the vehicle more than its metal counterparts, but these accessories are often expensive and extremely fragile, and therefore are vulnerable to damage or breakage. For example, carbon fiber vehicle accessories have been known to be damaged upon minimal contact with parking blocks and the like.

[0007] Given the increased market for vehicle accessories, accessories that are resistant to damage or breakage, but do not sacrifice performance and aesthetics of a vehicle in the process and can be obtained at lower prices are desired. Accordingly, the present invention is intended to provide a vehicle accessory made of aluminum composite material that is lightweight, durable and less

expensive, thus comprising a resilient yet functional accessory able to increase the performance and aesthetics of a vehicle.

[0008] The present disclosure provides a vehicle accessory for use with vehicles, such as automobiles. Specifically, the vehicle accessory is comprised of aluminum composite material, thereby decreasing the accessory's weight, without sacrificing resilience. The vehicle accessory is able to increase the performance of a vehicle by increasing the fuel efficiency of a vehicle, the comfort of a driver or passenger while driving, and the aerodynamic effects of a vehicle, while also being resistant to damage or breakage.

[0009] In an illustrative embodiment of the present disclosure, a vehicle accessory includes a front splitter configured to attach to the lower front portion of a vehicle. The front splitter comprises a body made substantially of aluminum composite material. The body includes a first portion, wherein holes may be disposed within the first portion, allowing for connection of the vehicle accessory to a bottom portion of a vehicle via a fixed connector. The body is also comprised of a second portion, which includes an end segment contoured to include a middle section that is substantially concave. The second portion end segment may be uniformly concave or may comprise of intermittent concave portions and convex portions. A protective rubber coating may be disposed on the end segment.

[0010] The illustrative front splitter further includes two connector types. First, the front splitter includes at least two adjustable length connectors illustratively spaced substantially equidistant with respect to the front splitter's second end (e.g., center axis). The adjustable length connectors illustratively include a first end for attachment to the front portion of a vehicle and a second end for attachment near the second end portion of the front splitter. The adjustable length connectors, when lengthened or shortened, are capable of raising or lowering the second end of the front splitter. Second, the front splitter illustratively includes at least one fixed connector, such as screws, bolts, push pin rivets, or the like.

[0011] The illustrative front splitter is made substantially of aluminum composite material. The aluminum composite material includes an upper layer of aluminum, disposed on the upper portion of a thermoplastic composite core. The aluminum composite material also includes a lower layer of layer of aluminum disposed on the lower portion of the thermoplastic composite core. The illustrative thermoplastic composite core, as mentioned above, is made substantially of polyethylene.

[0012] In another illustrative embodiment of the present disclosure, a vehicle accessory includes a canard configured to be selectively attached via at least one connector to the front portion of a vehicle. The canard includes a body made substantially of aluminum composite material. The body includes a first portion, which has a lip section wherein holes may be disposed throughout the lip section, allowing for connectors to attach the body of the vehicle accessory to the front portion (e.g., bumper) of a vehicle. The body also includes a second portion, extending outward from the first portion. The second portion is substantially concave, coming to a point at its end.

[0013] The illustrative canard further includes a connector extending through the holes that may be disposed within the first portion of the canard body, selectively attaching the canard body to the front portion of a vehicle.

[0014] The illustrative canard is made substantially of aluminum composite material. The aluminum composite material includes an upper layer of aluminum, disposed on the upper portion of a thermoplastic composite core. The aluminum composite material also includes a lower layer of layer of aluminum disposed on the lower portion of the thermoplastic composite core. The illustrative thermoplastic composite core, as mentioned above, is made substantially of polyethylene.

[0015] In yet another illustrative embodiment of the present disclosure, a vehicle accessory includes a side skirt configured to be selectively attached via the connector to the side bottom portion of a vehicle. The side skirt includes a body made substantially of aluminum composite

material. The body includes a first portion, wherein holes may be disposed throughout the first portion, allowing for the connectors to attach the body of the side skirt to the side bottom portion of a vehicle. The first portion may also include an edge section, wherein recesses are disposed within the edge section at uniform distances. The body also includes a second portion, extending outward from the first portion. The second portion is illustratively substantially linear.

[0016] The illustrative side skirt further includes at least one connector extending through the holes that may be disposed within the first portion of the side skirt body, selectively attaching the side skirt body to the side bottom portion of a vehicle.

[0017] The illustrative side skirt is made substantially of aluminum composite material. The aluminum composite material includes an upper layer of aluminum, disposed on the upper portion of a thermoplastic composite core. The aluminum composite material also includes a lower layer of layer of aluminum disposed on the lower portion of the thermoplastic composite core. The illustrative thermoplastic composite core, as mentioned above, is made substantially of polyethylene.

[0018] In yet another illustrative embodiment of the present disclosure, a vehicle accessory includes a diffuser configured to be selectively attached via the connector to a rearward bottom portion of a vehicle. The diffuser includes a body made substantially of aluminum composite material. The body is comprised of a first portion, wherein holes may be disposed throughout the first portion, allowing for the connectors to attach the body of the diffuser to the rearward bottom portion of a vehicle. The body also includes a second portion, extending from the first portion. The second portion may also include at least one fin unit, extending outward from the diffuser body.

[0019] The illustrative diffuser further includes at least one connector extending through the holes that may be disposed within the first portion of the diffuser body, selectively attaching the diffuser body to the rearward bottom portion of a vehicle.

[0020] The illustrative diffuser is made substantially of aluminum composite material. The aluminum composite material includes an upper layer of aluminum, disposed on the upper portion of a thermoplastic composite core. The aluminum composite material also includes a lower layer of layer of aluminum disposed on the lower portion of the thermoplastic composite core. The illustrative thermoplastic composite core, as mentioned above, is made substantially of polyethylene.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The above-mentioned and other features and advantages of this disclosure, and the manner of attaining them, will become more apparent and the invention itself will be better understood by reference to the following description of embodiments of the invention taken in conjunction with the accompanying drawings, wherein:

[0022] FIG. 1A is a disassembled perspective view of a first embodiment of a vehicle accessory of the present disclosure when disconnected from a vehicle, the vehicle accessory including a front splitter having a body, and at least one connector;

[0023] FIG. 1B is an assembled perspective view of the vehicle accessory of FIG. 1A, when connected to a vehicle;

[0024] FIG. 1C is a detail view of the assembled vehicle accessory of FIG. 1A, when connected to a vehicle;

[0025] FIG. 2 is a partial exploded perspective view of aluminum composite material found in the vehicle accessory embodiments of the present disclosure;

[0026] FIG. 3A is a disassembled perspective view of a second embodiment of a vehicle accessory of the present disclosure when disconnected from a vehicle, the vehicle accessory including a

canard having a body, and at least one connector;

[0027] FIG. 3B is an assembled perspective view of the vehicle accessory of FIG. 3A, when connected to a vehicle;

[0028] FIG. 3C is a detail view of the assembled vehicle accessory of FIG. 3A, when connected to a vehicle;

[0029] FIG. 4A is a disassembled perspective view of a third embodiment of a vehicle accessory of the present disclosure when disconnected to a vehicle, the vehicle accessory including a side skirt having a body, and at least one connector;

[0030] FIG. 4B is an assembled perspective view of the vehicle accessory of FIG. 4A, when connected to a vehicle;

[0031] FIG. 4C is another assembled perspective view of the vehicle accessory of FIG. 4A, when connected to a vehicle;

[0032] FIG. 5A is a perspective view of a fourth embodiment of a vehicle accessory of the present disclosure when connected to a vehicle, the vehicle accessory including a diffuser having a body, and at least one connector; and

[0033] FIG. 5B is an assembled profile view of the vehicle accessory of FIG. 5A, when connected to a vehicle.

DETAILED DESCRIPTION

[0034] The embodiments of the invention described herein are not intended to be exhaustive or to limit the invention to the precise form disclosed. Rather, the embodiments selective for description have been chosen to enable one skilled in the art to practice the invention.

[0035] Referring initially to FIGS. 1A-1C, a vehicle accessory 50 in the form of a front splitter is disclosed for use with a vehicle V, such as an automobile. The vehicle accessory 50 includes a body 52 which is comprised of a first or rear portion 54 and a second or front portion 56, an end segment 58, adjustable length connectors 60, and fixed connectors 61. Each element of vehicle accessory 50 is described further below.

[0036] As mentioned above, body 52 illustratively includes the first portion 54, and the second portion 56 longitudinally spaced relative to the first portion 54. The first portion 54 is disposed substantially underneath vehicle V and represents approximately 67% of the vehicle accessory body 52. Holes 62 may be disposed throughout the first portion 54, so as to allow fixed connectors 61 to pass through the first portion 54. Fixed connectors 61 may also pass through holes in the underside of vehicle V, thereby selectively attaching first portion 54 to vehicle V. In the illustrative embodiment shown, fixed connectors 62 represent push lock rivets. In other embodiments fixed connectors 61 may be of a different style of connector, such as a screw, bolt, fastener, or the like.

[0037] Second portion 56 extends longitudinally outward (forward) from first portion 54 and makes up approximately 33% of the vehicle accessory body 52. Second portion 56 is partly disposed underneath vehicle V, and partly extends out from underneath vehicle V. Second portion 56 also includes a front end segment or edge 58. End segment 58 is contoured so as to include a middle or center section 65 that is substantially concave. The end segment 58 of second portion 56 may be uniformly concave or may be comprised of intermittent concave portions and convex portions. A protective rubber coating or trim may cover end segment 58.

[0038] The adjustable length connectors 60 are configured to attach the second portion 56 to the front portion of the vehicle V. Each adjustable length connector 60 illustratively includes a first end 66 and a second end 68. First end 66 is disposed through holes 69 in the front portion of vehicle V. Second end 68 is disposed through holes 71 in the second portion 56 of vehicle accessory body 52. The adjustable length connector 60 may be lengthened or shortened (illustratively through a threaded coupler 73), thereby capable of raising or lowering the second portion 56 of the vehicle accessory of vehicle accessory body 52.

[0039] Referring to FIGS. 3A-3C, another embodiment of the vehicle accessory 150 in the form of a canard is disclosed. The vehicle accessory 150 has similarities in both the structure and operation

as vehicle accessory **50**. Like elements of the vehicle accessory **50** are identified by adding “100” to the corresponding reference number of the vehicle accessory **50**.

[0040] Vehicle accessory **150** illustratively includes a body **152** having a first portion **154** and a second portion **156**, a lip segment or edge **158**, and fixed connectors **161**. Each element of vehicle accessory **150** is described further below.

[0041] As mentioned above, the body **152** illustratively includes the first or rear portion **154** and the second or front portion **156** longitudinally spaced relative to the first portion **154**. The first portion **154** is disposed nearest to the front portion of vehicle V and represents approximately 50% of the vehicle accessory body **152**. The first portion **154** includes lip segment **158**, wherein holes **162** may be disposed throughout lip segment **158** of first portion **154**, so as to allow fixed connectors **161** to pass through the first portion **154**. Fixed connectors **161** may also pass through holes in the front portion of vehicle V, thereby selectively attaching first portion **154** to vehicle V. In the illustrative embodiment shown, fixed connectors **161** represent push lock rivets. In other embodiments fixed connectors **161** may be of a different style of connector, such as a screw, bolt, fastener, or the like.

[0042] The body **152** of the vehicle accessory **150** also illustratively includes second portion **156**, extending longitudinally outward (forward) from the first portion **154**. The second portion **156** is substantially convex, coming to a point at an end segment **155**. In the illustrative embodiment shown, body **152** forms a trapezoidal shape. In other embodiments, the body **152** may form a number of different shapes.

[0043] Referring now to FIGS. 4A-4C, a vehicle accessory **250** in the form of a side skirt has similarities in both the structure and operation as vehicle accessory **50**. Like elements of the vehicle accessory **50** are identified by adding “200” to the corresponding reference number of the vehicle accessory **50**.

[0044] Vehicle accessory **250** illustratively includes a body **252** which is comprised of a first or inner portion **254** and a second or outer portion **256**, an edge section **258**, and fixed connectors **262**. Each element of vehicle accessory **250** is described further below.

[0045] As mentioned above, illustrative body **252** of the vehicle accessory **250** includes the first portion **254** and the second portion **256** laterally spaced from the first portion **254**. The first portion **254** is disposed underneath the side bottom portion of vehicle V and represents approximately 50% of the vehicle accessory body **252**. Fixed connectors **261** may be disposed through holes made in the first portion **254**. Fixed connectors **261** may also pass through holes in the underside of the side bottom portion of vehicle V, thereby selectively attaching first portion **254** to vehicle V. In the illustrative embodiment shown fixed connectors **261** represent push lock rivets. In other embodiments fixed connectors **261** may be of a different style of connector, such as a screw, bolt, fastener, or the like. The first portion may also include edge section **258**, wherein inwardly extending recesses are illustratively disposed within edge section **258**. In the illustrative embodiment shown, the recesses of edge section **258**, are at uniform distances apart. In other embodiments, the recesses of edge section **258**, may be at variable distances apart.

[0046] The body **252** of vehicle accessory **250** is also comprised of a second portion **256**, extending laterally outward from the first portion **254**. Second portion **256** protrudes out from the side of the vehicle V (and is thus not underneath it). Second portion **256** represents approximately 50% of the vehicle accessory body **252**. In the illustrative embodiment shown, body **252** forms a substantially rectangular shape. In other embodiments, the body **252** may form a number of shapes. Second portion **256** may also include a fin **253** extending upward from the second portion **256**.

[0047] Referring now to FIGS. 5A and 5B, a vehicle accessory **350** in the form of a diffuser has similarities in both the structure and operation as vehicle accessory **50**. Like elements of the vehicle accessory **50** are identified by adding “300” to the corresponding reference number of the vehicle accessory **50**.

[0048] Vehicle accessory **350** illustratively includes a body **352** which is comprised of a first or

front portion **354** and a second or rear portion **356**, at least one fin unit **358**, and fixed connectors **362**. Each element of vehicle accessory **350** is described further below.

[0049] As mentioned above, illustrative body **352** of vehicle accessory **350** includes the first portion **354**, and the second portion **356** longitudinally spaced relative to the first portion **354**. The first portion **354** is disposed underneath the rearward bottom portion of vehicle V and represents approximately 67% of the vehicle accessory body **352**. The first portion **354** may be contoured so as to coordinate with the rearward portion of the vehicle that it is disposed against. Fixed connectors **362** may be disposed through holes made in the first portion **354**. Fixed connectors **362** may also pass through holes in the underside of the rearward bottom portion of vehicle V, thereby selectively attaching first portion **354** to vehicle V. In the illustrative embodiment shown fixed connectors **362** represent push lock rivets. In other embodiments fixed connectors **362** may be of a different style of connector, such as a screw, bolt, fastener, or the like.

[0050] Second portion **356** illustratively extends longitudinally outward (forward) from first portion **354** and makes up approximately 33% of the vehicle accessory body **352**. Second portion **356** is partly disposed underneath vehicle V, and partly protrudes out from the rear of the vehicle V (and is thus not underneath it). Second portion **356** includes at least one fin unit **358**. Fin unit **358** may extend either upward or downward from the second portion **356** of the body **352** of vehicle accessory **350**.

[0051] Referring now to FIG. 2, the vehicles accessory of all embodiments detailed above (**50**, **150**, **250** and **350**) are made substantially (60% or more) of aluminum composite material **70**. More particularly, the aluminum composite material of each illustrative body **52**, **152**, **252** and **352** is comprised of an upper layer of aluminum (or aluminum alloy) **72**, disposed on the upper portion or surface **74** of a thermoplastic composite core **76**. The aluminum composite material also includes a lower layer of layer of aluminum (or aluminum alloy) **78** disposed on the lower portion or surface **80** of the thermoplastic composite core **76**. In other words, the thermoplastic composite core **76** is sandwiched between the upper layer of aluminum **72** and the lower layer of aluminum **78**.

[0052] Illustratively, the upper layer **72** and the lower layer **78** are both formed of aluminum or an aluminum alloy. The thermoplastic composite core **76**, as mentioned above, is made substantially (60% or more) of polyethylene. In an illustrative embodiment, the aluminum composite material **70** material ranges between 0.5 mm and 4.0 mm in thickness. In certain illustrative embodiments, each aluminum layer **72** and **78** has a thickness of between 0.05 mm and 0.1 mm. As such, the intermediate thermoplastic composite core **76** illustratively has a thickness of between 0.1 mm and 3.9 mm.

[0053] Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the spirit and scope of the invention as described and defined in the following claims.

Claims

1.-15. (canceled)

16. A vehicle accessory comprising: a front splitter, the front splitter comprising a front edge and a rear edge, the front edge being curved along at least a portion thereof, and the front splitter being made substantially of an aluminum composite material, the aluminum composite material comprising a first layer, a second layer, and a thermoplastic core disposed between the first layer and the second layer, wherein the front edge comprises intermittent concave and convex portions, each having a thickness the same as the thickness of the front splitter measured orthogonally through the center of mass of the front splitter.

17. The vehicle accessory of claim 16, wherein the aluminum composite material has a thickness of between 0.5 mm and 4.0 mm.

18. The vehicle accessory of claim 17, wherein the thickness of the intermittent concave and

convex portions is the same as the thickness of between 0.5 mm and 4.0 mm of the aluminum composite material.

19. The vehicle accessory of claim 16, wherein the concave and convex portions comprise a first convex portion centrally disposed along the front edge, a first concave portion extending from the first convex portion, a second concave portion extending from the first convex portion and being laterally opposed from the first concave portion.

20. The vehicle accessory of claim 19, wherein the concave and convex portions further comprise a second convex portion extending between the first concave portion and the rear edge and a third convex portion extending between the second concave portion and the rear edge.

21. The vehicle accessory of claim 16, further comprising a connector for attaching the front splitter to a vehicle.

22. The vehicle accessory of claim 21, wherein the connector is configured to raise or lower the front edge of the front splitter in response to an adjustment of the connector.

23. The vehicle accessory of claim 22, wherein the adjustment comprises lengthening or shortening the connector.

24. A vehicle accessory comprising: a front splitter, the front splitter comprising a front edge and a rear edge, the front edge being curved along at least a portion thereof, and the front splitter being made substantially of an aluminum composite material, the aluminum composite material comprising a first layer, a second layer, and a thermoplastic core disposed between the first layer and the second layer, wherein the aluminum composite material has a thickness of between 0.5 mm and 4.0 mm, wherein the front edge comprises intermittent concave and convex portions, each having a thickness the same as the thickness of between 0.5 mm and 4.0 mm of the aluminum composite material.

25. The vehicle accessory of claim 24, wherein the thickness of the intermittent concave and convex portions is the same as the thickness of the front splitter measured orthogonally through the center of mass of the front splitter.

26. The vehicle accessory of claim 24, wherein the concave and convex portions comprise a first convex portion centrally disposed along the front edge, a first concave portion extending from the first convex portion, a second concave portion extending from the first convex portion and being laterally opposed from the first concave portion.

27. The vehicle accessory of claim 26, wherein the concave and convex portions further comprise a second convex portion extending between the first concave portion and the rear edge and a third convex portion extending between the second concave portion and the rear edge.

28. The vehicle accessory of claim 24, further comprising a connector for attaching the front splitter to a vehicle.

29. The vehicle accessory of claim 28, wherein the connector is configured to raise or lower the front edge of the front splitter in response to an adjustment of the connector.

30. The vehicle accessory of claim 29, wherein the adjustment comprises lengthening or shortening the connector.

31. A vehicle accessory comprising: a front splitter, the front splitter comprising a front edge and a rear edge, the front edge comprising intermittent concave and convex portions, and the front splitter being made substantially of an aluminum composite material, the aluminum composite material comprising a first layer, a second layer, and a thermoplastic core disposed between the first layer and the second layer, wherein the intermittent concave and convex portions each have a thickness the same as a thickness of the aluminum composite material.

32. The vehicle accessory of claim 31, wherein the aluminum composite material has a thickness of between 0.5 mm and 4.0 mm.

33. The vehicle accessory of claim 31, wherein the concave and convex portions comprise a first convex portion centrally disposed along the front edge, a first concave portion extending from the first convex portion, a second concave portion extending from the first convex portion and being

laterally opposed from the first concave portion, a second convex portion extending between the first concave portion and the rear edge, and a third convex portion extending between the second concave portion and the rear edge.

34. The vehicle accessory of claim 31, further comprising a connector for attaching the front splitter to a vehicle.

35. The vehicle accessory of claim 34, wherein the connector is configured to raise or lower the front edge of the front splitter in response to a lengthening or shortening of the connector.
